Programming in Finance II: Big Projects 2026

Final version

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General. The projects are deliberately formulated in an open way. A detailed formulation of the problem and a project plan **with presentation** is already part of the project. This includes the decision, which features to implement beyond the minimum requirements and how to document your project (beyond the mandatory academic project documentation).

Goal. The main goal is to produce one working solution. Complexity is rewarded, but everything that you hand in has to work.

Process over Product on Github. In the age of AI, anyobdy can create a decent product. The challenge is to communicate the requirements, to organize the process and to provide useful feedback to AI. The entire development process must happen on GitHub. This provides a transparent record of your work via the commit history. Commit regularly to demonstrate progress, with meaningful commit messages that reflect the evolution of the project.

Agentic Project. Organize your project as an agentic project. This means that AI agents should be able to contribute in a meaningful way to your project. Provide as minimum an `AGENTS.md` file.

Presentation

– Present your project plan in a 15x3 group presentation.
– Six minutes per group
– You decide who speaks

Delivery via Github

– Publish your project on Github. You decide on the specific format.
– Include all steps for installation and deployment in the documentation.
– **Provide a** user guide
– Minimum documentation requirements: Comprehensive README.md with project overview, installation instructions, usage examples, and API documentation; commit history showing regular progress; issues or project board for tracking tasks; and a wiki or docs folder for additional technical details.

Academic documentation.

– Deliver the project documentation in PDF on iCorsi (5-8 pages pure text)
– Use LaTeX
– Link to your Github page
– Project plan
– Project diary (your steps, what you did, why, which tools)
– Economics/mathematics/data science behind your project
– Sample results or applications and reflection of (economic) results
– Lessons learned from the project

Criteria

1. Problem definition
   Creative? Useful? New?
2. Problem solution
   Complete? Works? Realistic results? Suitable algorithms/methods?
3. Scope
   Did you do the work? Difficulty/complexity/comprehensiveness
4. Coding: structure, style
5. Regular use of GitHub and AI agents. At least one pull request made by an AI agent. **Agentic project organization.**
6. Quality and depth of the academic documentation

Material from the web or from other courses

Use of foreign code (libraries, tutorials from stackexchange or others) is permitted. Foreign code (=anything that you found on the web and that you did not write yourself for this project) needs to be cited correctly. Foreign code is ignored when assessing your contribution (i.e. it counts neither positively nor negatively towards the contribution. This implies that if you used only foreign code, there is no contribution from your part).

It is absolutely forbidden to use one project in two courses. If you work on similar problems in two courses, make sure that your project is a safe distance away from anything you or your colleagues are doing any other course.

Use of generative AI

Use of generative AI is expected and should be acknowledged (which tools?) in the PDF documentation.

1 Generic project criteria

In 2026, the rule for projects is just this: Combine at least **three** of the following elements in your project, with at least one being an advanced component **(denoted with *)**.

- Cloud-based infrastructure **any provider, minimum is using a VPS**
- ***Use of a large language model (can be locally or via any API)**
- *Machine learning models (e.g., reinforcement learning for trading, NLP for sentiment analysis)
- *Blockchain integration (e.g., DeFi protocols, smart contracts)
- **A non-trivial database: SQL with multiple tables, vector stores, other non-trivial solutions**
- **Real-time data processing (i.e. live market data)**
- Advanced data visualization (e.g., interactive dashboards with AI insights)
- Creating your own API with either authentication or rate limits
- Cybersecurity measures (e.g., encryption, secure APIs, login systems)
- **Web frontend with a focus on user experience**
- Mobile/web app with advanced UX
- A meaningful open source contribution, e.g. publishing a useful Python Package

2 2026 Project Ideas

These projects build on the 2025 inspirations but incorporate cutting-edge technologies and AI/ML to reflect the advanced state of AI coding in 2026. Projects must demonstrate deep understanding of financial concepts while leveraging modern programming techniques.

2.1 AI-Powered Robo-Advisor Platform

Minimum requirements: Develop a robo-advisor platform that uses machine learning to provide personalized investment advice. Implement portfolio optimization, integrate market data, and create a web dashboard for user interaction. **Nice to have:** risk assessment based on user profiles and market conditions, using an LLM for financial advice generation.

2.2 DeFi Yield Farming Optimizer

Minimum requirements: Build a system that analyzes DeFi protocols across multiple blockchains to optimize yield farming strategies. Use machine learning to predict APY changes, implement automated rebalancing via smart contracts, and provide a real-time dashboard. Include risk metrics and impermanent loss calculations. **Nice to have:** Multi-chain support, integration with local LLMs for strategy explanations, advanced Monte Carlo simulations for risk modeling.

2.3 Cryptocurrency Arbitrage Trading Bot

Minimum requirements: Create an arbitrage trading bot that operates across multiple exchanges and blockchains. Implement real-time price monitoring and automatic detection of arbitrage opportunities. **Nice to have:** execute trades via APIs, proper error handling and risk controls. Backtesting capabilities and performance analytics.

2.4 ESG Analytics of corporate reports

Minimum requirements: Build a platform that uses NLP/LLMs to analyze ESG (Environmental, Social, Governance) factors for investment decisions. Obtain data either from EDGAR or scrap for annual reports from company websites. Implement a model for calculating your own ESG scores. **Nice to have:** interactive visualizations for portfolio managers, report creation with LLMs, monitoring of news/social media.

2.5 LLMs and RAG

Minimum requirements: Similar to the ESG project, obtain a large corpus of financial texts (e.g., from EDGAR, news articles, research papers) about **many (100+) companies**. Find an interesting question that can be answered with RAG (Retrieval Augmented Generation) and analyze the question for each company in the corpus in a *systematic* way and produce systematic output. **Nice to have:** visualize the results.

2.6 Algorithmic Trading

Minimum requirements: Implement a trading agent for the equity, FX or crypto market and integrate with live market data streams. Support at least three assets. Implement a portfolio optimization framework. **Nice to have:** use a simple ML model for predicting price movements, implement a web dashboard, implement simple risk management, use an LLM for generating trading insights.

2.7 AI-Enhanced Options Pricing Platform

Minimum requirements: Build a platform for advanced options pricing. Implement volatility surface modeling and calculation for all Greeks. Calculate options prices with one of these advanced methods: monte carlo simulations, binomial trees, or FFT-based methods. **Nice to have:** use an LLM for explaining the pricing results, implement a web dashboard, visualize the volatility surface.

2.8 Backtesting Framework for Quantitative Strategies

Minimum requirements: Create a backtesting framework that supports multiple asset classes (equities, FX, crypto) and allows users to test quantitative trading strategies. Implement performance metrics and risk analysis tools. **Nice to have:** integration with market data for paper trading, possibility for the user to define custom strategies, web dashboard for visualizing backtest results.

2.9 Your custom project

Decide on your own topic. Satisfy the generic project criteria (1). For your own topic, present your detailed project plan at the first presentation date (April 22).